HR ANALYTICS DASHBOARD – PROJECT REPORT

Introduction

Human Resource Analytics (HR Analytics) helps organizations analyze workforce-related data to improve employee management, reduce attrition, and support strategic decision-making.

This project focuses on building an interactive HR Analytics Dashboard in Tableau to analyze employee attrition patterns, demographic distribution, job satisfaction, and departmental trends.

The dashboard provides HR managers with a clear visual understanding of employee turnover and helps identify major factors contributing to attrition.

Objective of the Project

The main objectives of this project are:

To monitor employee attrition across departments
To identify attrition trends by age, gender, and education
To analyze job satisfaction levels among employees
To support HR decision-making through interactive visualizations
To provide a centralized dashboard for workforce analysis

Tools and Technologies Used

Tableau – Dashboard development and visualization
Microsoft Excel / CSV Dataset – Data preprocessing
HR Employee Dataset – Employee demographic and organizational data

Dataset Description

The dataset used in this project contains detailed information about employees in an organization. It is used to analyze workforce trends and employee attrition.

Key Attributes in Dataset:

- **Employee ID** – Unique identifier for each employee
- **Age** – Age of employees
- **Gender** – Male/Female classification

- **Education** – Educational qualification of employees
- **Job Role** – Position or designation of employees
- **Salary** – Monthly income of employees
- **Years at Company** – Duration of employment
- **Attrition** – Indicates whether the employee has left the company

Dataset Details:

- Total Records: **196 employees**
- Contains both **numerical and categorical data**
- Used for analyzing trends like attrition rate, salary distribution, and employee demographics

The dataset includes 1,470 employee records.

Key Performance Indicators (KPIs) Used in Dashboard

The dashboard highlights major HR metrics:

Employee Count: 1,470

Attrition Count: 237

Attrition Rate: 16.12%

Active Employees: 1,233

Average Age: 37 years

These KPIs help quickly assess overall workforce status.

Dashboard Components and Analysis

A. Department-wise Attrition

The pie chart shows attrition distribution across departments:

Sales has highest attrition

R&D has moderate attrition

HR has lowest attrition

This helps identify departments with higher employee turnover.

B. Employee Distribution by Age Group

The histogram displays employee count by age:

Highest employee concentration between age 30–35

Lower workforce representation above age 50

This helps understand workforce age composition.

C. Attrition by Gender

The dashboard compares male and female attrition:

Male attrition count is higher than female attrition

This can support gender-focused retention strategies.

D. Job Satisfaction Rating Analysis

A heatmap shows job satisfaction across different job roles.

Insights:

Certain job roles show lower satisfaction levels
Roles with low satisfaction may require HR intervention

E. Education Field-wise Attrition

Bar chart indicates attrition by education field:

Highest attrition observed in:

Life Sciences
Medical
Marketing

This helps analyze educational background trends in attrition.

F. Attrition Rate by Gender across Age Groups

Donut charts compare male and female attrition in age categories:

Under 25
25–34
35–44
45–54
Over 55

The age group 25–34 shows highest attrition.

Insights Derived from Dashboard

Major insights observed:

Attrition is highest in Sales department
Employees aged 25–34 leave more frequently
Male attrition is comparatively higher
Life Sciences field contributes highest attrition
Job satisfaction varies significantly across job roles

- ## Benefits of the Dashboard

This dashboard helps HR teams:

Track workforce health in real time
Identify attrition risk areas
Improve retention strategies
Support data-driven HR decisions

Conclusion

The HR Analytics Dashboard successfully converts raw employee data into meaningful business insights using Tableau. It helps organizations understand attrition behavior and supports better workforce planning.

This project demonstrates how data visualization can improve HR decision-making through interactive dashboards.